

Sturm-Liouville Problems:

Consider a second order ODE of the form

$$[p(x)y']' + [q(x) + \lambda r(x)]y = 0 \quad \dots\dots (1)$$

on some interval $a \leq x \leq b$, satisfying conditions of the form

$$(2) \quad \dots \left\{ \begin{array}{l} k_1 y + k_2 y' = 0 \quad \text{at } x=a \\ l_1 y + l_2 y' = 0 \quad \text{at } x=b \end{array} \right. \dots 2(a) \quad \dots 2(b)$$

Here λ is a parameter and k_1, k_2, l_1, l_2 are given real constants. Furthermore, at least one of each constant in each condition of (2) must be different from zero.

Equation (1) is called Sturm-Liouville equation.

Together with 2(a) and 2(b), it is known as Sturm-Liouville Problem.

It is an example of Boundary Value

Problem.

Note: A boundary value problem consists of an ODE and given boundary conditions referring to the two boundary points (end points) $x=a$ and $x=b$ of a given interval $[a, b]$

Eigen Values, Eigen Functions:

clearly $y \equiv 0$ is a solution of the problem ① (Trivial solution)

for any λ .

Would like to find eigenfunctions $y(x)$, that is solutions of ① satisfying ② without being identically zero.

We call a number λ for which an eigen function exists an eigen value of the Sturm-Liouville equations.

Note:

(1) Under rather general conditions on the

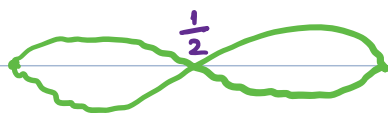
functions p, q, r in (1), the Sturm-Liouville problem (1), (2) has infinitely many eigenvalues.

(2) If p, q, r and p' in (1) are real-valued and continuous on $[a, b]$ and r is positive throughout the interval (or negative throughout the interval), then all eigenvalues of the Sturm-Liouville problem (1), (2) are real.

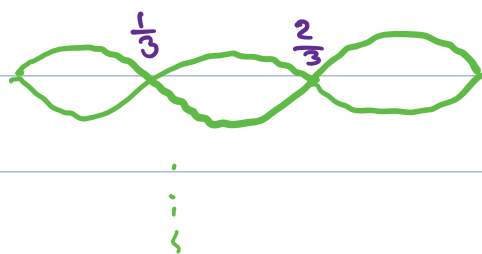
Ex: Vibrating String:



$$y'' + \lambda y = 0, \quad y(0) = 0, \quad y(\pi) = 0$$



We can reformulate the problem as a Sturm-Liouville problem as:



$$(1y')' + (0 + \lambda 1)y = 0$$

$$1y(0) + 0y'(0) = 0$$

$$1y(\pi) + 0y'(\pi) = 0$$

(Here $v(x) \equiv 1$)

• If $\lambda = -v^2$ is negative, the general solution is
$$y = c_1 e^{vx} + c_2 e^{-vx}$$

From the boundary conditions, we get $c_1 = c_2 = 0$

If $\lambda = 0$, the general solution is
$$y = (c_1 + c_2 x)$$

and again from the boundary conditions
 $c_1 = c_2 = 0$

If $\lambda = v^2$ is positive, the general solution is

$$y = c_1 \cos(vx) + c_2 \sin(vx)$$

From the boundary conditions $c_1 = 0$

$$\text{and } y(\pi) = c_2 \sin(v\pi) = 0$$

$$\Rightarrow v = \pm 1, \pm 2, \dots$$

Therefore the functions

$$y_v = \sin(vx), \quad v = \sqrt{\lambda} = 1, 2, 3, \dots$$

are eigen functions of the ODE and their associated eigenvalue is $\lambda = v^2$

Note that the solution of the problem is precisely the Trigonometric system of the Fourier series.

Orthogonal Functions

Orthogonality:

Let us define the inner product of two functions y_m and y_n with respect to the weight function $r(x)$ ($r(x) > 0$) in the interval $[a, b]$ as :

$$\langle f, g \rangle_r = \int_a^b r(x) f(x) g(x) dx$$

The norm of a function is

$$\|f\|_r = \sqrt{\langle f, f \rangle_r}$$

Two functions f and g are called

orthogonal if $\langle f, g \rangle_r = 0$

(with respect to the weight function w)

A set of functions $\{y_1, y_2, \dots\}$

is called orthonormal if

$$\left\{ \begin{aligned} \langle y_m, y_n \rangle_x &= \delta_{mn} = 0 \text{ if } m \neq n \\ &= 1 \text{ if } m = n \end{aligned} \right\}$$

Ex: (continued from previous example):

The set of functions $y_v = \sin(vx)$

where $v = \sqrt{\lambda} = 1, 2, \dots, 3$

are orthogonal in the interval $[0, \pi]$.

$$\begin{aligned} \langle y_{v_1}, y_{v_2} \rangle &= \int_0^\pi \sin(v_1 x) \sin(v_2 x) dx \\ &= \frac{1}{2} \int_0^\pi \cos[(v_1 - v_2)x] dx - \frac{1}{2} \int_0^\pi \cos[(v_1 + v_2)x] dx \\ &= \frac{1}{2} \frac{\sin[(v_1 - v_2)x]}{(v_1 - v_2)} \Big|_0^\pi - \frac{1}{2} \frac{\sin[(v_1 + v_2)x]}{(v_1 + v_2)} \Big|_0^\pi \\ &= 0 \end{aligned}$$

but they are not orthonormal because

$$\|y_v\|^2 = \int_0^\pi \sin^2(vx) dx = \int_0^\pi \left[\frac{1 - \cos(2vx)}{2} \right] dx$$

$$= \frac{\pi}{2}$$

The set of functions $\tilde{y}_\nu = \sqrt{\frac{2}{\pi}} \sin(\nu x)$
 $\nu = \sqrt{\lambda} = 1, 2, 3, \dots$

is orthonormal.

Orthogonality of Eigen functions in case of Sturm-Liouville problems:

If p, q, r and p' are real valued and continuous in the interval $[a, b]$ and $r > 0$.

Let the function y_m and y_n be eigenfunctions associated to different eigenvalues λ_m and λ_n , then

$$\langle y_m, y_n \rangle_r = 0$$

(i.e. y_m and y_n are orthogonal on $[a, b]$ with respect to the weight function r)

- If $p(a)=0$, then 2(a) can be dropped from the problem.

If $p(b)=0$, then 2(b) can be dropped from the problem.

It is then required that y and y' remain bounded at such a point and the problem is called singular as opposed to a regular problem in which (2) is used.

- If $p(a)=p(b)$ then (2) can be replaced by the "periodic boundary conditions"

$$\left. y(a)=y(b), y'(a)=y'(b) \right\} \dots (3)$$

- The boundary value problem consisting of the Sturm-Liouville equation (1) and the periodic boundary condition is called periodic-Sturm-Liouville problem.

Mixed Dirichlet - Neumann boundary Condition :

$$k_1 y(a) + k_2 y'(a) = \alpha$$

$$l_1 y(b) + l_2 y'(b) = \beta$$

If $\alpha = \beta = 0$, the boundary conditions are said to be homogeneous.

If $k_2 = l_2 = 0$, they are called Dirichlet boundary conditions.

If $k_1 = l_1 = 0$, they are called Neumann boundary conditions.

Singular Sturm - Liouville problem

A Sturm-Liouville problem

$$(p(x)y')' + [q(x) + \lambda r(x)]y = 0$$

is called singular in any one of the following cases.

- $p(a) = 0$, Boundary Condition ^{at a} is dropped, Boundary condition at b is homogeneous mixed.

- $p(b) \neq 0$, Boundary condition at b is dropped, Boundary Condition at a is homogeneous, mixed.
- $p(a) = p(b) = 0$ and there is no Boundary Condition.
- The interval $[a, b]$ is infinite. Otherwise the problem is called regular.

Ex: Legendre's Equation and Polynomials.

Legendre Equation:

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0$$

is a Sturm-Liouville problem

$$\left[(1-x^2)y' \right]' + n(n+1)y = 0 \quad [-1, 1].$$

ie. of the form $\left[p(x)y' \right]' + [q(x) + \lambda r(x)]y = 0$

with $p = 1-x^2$, $q=0$, $r=1$

Note that

$p(-1) = p(1) = 0$ and so the

Sturm-Liouville problem is singular and we do not need boundary conditions.

- The Legendre polynomial $P_n(x)$ is a non-trivial solution of the problem associated to the eigen value $\lambda = n(n+1)$
- By the previous theorem, Legendre polynomials are orthogonal in the interval $[-1, 1]$.

Generalized Fourier Series

Let the set $\{y_1, y_2, \dots\}$ be orthogonal with respect to the weight function r in an interval $[a, b]$.

Let f be a function that we want to expand in this orthogonal basis

$$f = \sum_{m=0}^{\infty} a_m y_m(x)$$

To find the Fourier Coefficients a_m , we compute the inner product of f with y_n .

$$\langle f, y_n \rangle_r = \left\langle \sum_{m=0}^{\infty} a_m y_m(x), y_n \right\rangle_r$$

$$= \sum_{m=0}^{\infty} a_m \langle y_m(x), y_n(x) \rangle_r$$

$$= a_n \|y_n\|_r^2$$

$$\text{So, } a_n = \frac{\langle f, y_n \rangle_r}{\|y_n\|_r^2} = \frac{\int_a^b r f y_n dx}{\int_a^b r y_n^2 dx}$$

Fourier-Legendre Series

Legendre polynomials $P_m(x)$ are orthogonal in $[-1, 1]$ with respect to $r(x)=1$.

In this interval we can perform an eigen function expansion of the form

$$f = \sum_{m=0}^{\infty} \frac{\langle f, P_m \rangle}{\|P_m\|^2} P_m(x)$$

It can be shown that

$$\|P_m\|^2 = \frac{2}{2m+1}$$

Ex: $f = \sin(\pi x)$

$$a_m = \frac{2m+1}{2} \int_{-1}^1 \sin(\pi x) P_m(x) dx$$

Then $f = .95493 P_1 - 1.15824 P_3$
 $+ .21929 P_5 - .01664 P_7$
 $+ .00068 P_9 - .00002 P_{11} + \dots$

- Similarly look at Fourier-Bessel series.

- Ex: Expand $f(x) = \frac{1}{1-x^2}$ in terms of Fourier-Bessel series

Ex: Find the eigen values and eigen functions.

Verify orthogonality.

$$y'' - 2y' + (\lambda + 1)y = 0 \quad ; \quad y(0) = 0, \quad y(1) = 0$$

Note: Any equation of the form $y'' + f y' + (g + \lambda h)y = 0$

can be transformed to $(p y')' + (q + \lambda r)y = 0$

$$\text{with } p = e^{\int f dx}, \quad q = p g, \quad r = p h$$

In the above problem $p = e^{\int -2 dx} = e^{-2x}$

$$q = p \cdot 1 = p = e^{-2x}, \quad r = p \cdot 1 = p = e^{-2x}$$

$$\text{Then } (p y')' = (e^{-2x} y')' = y'' e^{-2x} - 2 e^{-2x} y'$$

$$(q + \lambda r)y = (p + p y) = e^{-2x}(\lambda + 1)y$$

$$\text{So, } (p y')' + (q + \lambda r)y = e^{-2x}(y'' - 2y' + (\lambda + 1)y) = 0$$

Here the weight function $r(x) = e^{-2x}$

Now for the solution of the differential equation, the auxiliary equation is $m^2 - 2m + (\lambda + 1) = 0$

$$\Rightarrow m = \frac{2 \pm \sqrt{4 - 4(\lambda + 1)}}{2} = 1 \pm \sqrt{-\lambda}$$

Case 1: $\lambda > 0$ let $\sqrt{\lambda} = \mu$

$$\text{Then } m = 1 \pm i\mu$$

So, the solution is $y(x) = e^x (C_1 \cos(\mu x) + C_2 \sin(\mu x))$

Now $y(0)=0 \Rightarrow c_1=0$; Then $y(x)=c_2 e^x \sin(\mu x)$

$$y(1)=0 \Rightarrow c_2 e^1 \sin(\mu) = 0$$

For nontrivial solution $c_2 \neq 0$

$$\text{Then } \sin \mu = 0 \Rightarrow \mu = n\pi, \quad n = \pm 1, \pm 2, \dots$$

The eigen values of $\lambda_n = n^2 \pi^2, \quad n = 1, 2, 3, \dots$

The corresponding eigen functions are $y_n(x) = e^x \sin(n\pi x)$

Orthogonality: $w(x) = e^{-2x}$

$$\text{Now } \int_0^1 y_n(x) y_m(x) e^{-2x} dx$$

$$= \int_0^1 e^x \sin(n\pi x) e^x \sin(m\pi x) e^{-2x} dx$$

$$= \int_0^1 \sin(n\pi x) \sin(m\pi x) dx$$

$$= \frac{1}{2} \int_0^1 [\cos\{(n-m)\pi x\} - \cos\{(n+m)\pi x\}] dx$$

$$= \frac{1}{2} \left[\frac{\sin\{(n-m)\pi x\}}{(n-m)\pi} - \frac{\sin\{(n+m)\pi x\}}{(n+m)\pi} \right]_0^1$$

$$= 0 \quad \text{if } n \neq m$$

$$= \int_0^1 \sin^2(n\pi x) dx = \int_0^1 \frac{[1 - \cos(2n\pi x)]}{2} dx$$

$$= \left[\frac{x}{2} - \frac{\sin(2n\pi x)}{4n\pi} \right]_0^1 = \frac{1}{2} \neq 0$$

Thus orthogonality of eigen functions has been shown.

Case 2: $\lambda \leq 0$ Let $\lambda = -\alpha^2$, $\alpha > 0$

Then $r = 1 \pm \alpha$

$$\text{Then } y(x) = c_1 e^{(1+\alpha)x} + c_2 e^{(1-\alpha)x}$$

$$y(0) = 0 \Rightarrow c_1 + c_2 = 0 \Rightarrow c_1 = -c_2$$

$$\text{Then } y(x) = c_1 \left[e^{(1+\alpha)x} - e^{(1-\alpha)x} \right]$$

$$\text{Now } y(1) = 0 \Rightarrow c_1 (e^{1+\alpha} - e^{1-\alpha}) = 0$$

For nontrivial solution $c_1 \neq 0$

$$\text{Then } e^{1+\alpha} = e^{1-\alpha} \Rightarrow e^{2\alpha} = 1 \Rightarrow \alpha = 0$$

$$\text{Hence } \lambda = 0 \text{ and so } y(x) = c_1 (e^x - e^x) = 0$$

$\Rightarrow y(x) \equiv 0$ and no nontrivial solution exists.

Fourier Transform:

If f is a absolutely integrable (i.e. $\int_{-\infty}^{\infty} |f| dx < \infty$) on $\mathbb{R}$ and piecewise continuous on every finite interval then the Fourier Transform of f is defined by

$$\hat{f}(s) = \mathcal{F}(f)(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ist} f(t) dt$$

(Notation)

$$\text{. Now } \mathcal{F}(f)(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ist} f(t) dt$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} [\cos(st) - i \sin(st)] f(t) dt$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \cos(st) f(t) dt - i \int_{-\infty}^{\infty} \sin(st) f(t) dt$$

(Fourier Cosine Transform)

(Fourier Sine Transform)

Inverse Fourier Transform:

$$\text{Suppose } \mathcal{F}(f)(s) = \hat{f}(s)$$

Then
$$\hat{f}(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ist} f(t) dt$$

Then
$$f(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(s) e^{ist} ds$$

This is called the inverse Fourier Transform of f i.e. $\mathcal{F}^{-1}(\hat{f}) = f$
if $\mathcal{F}(f) = \hat{f}$

Existence of the Fourier Transform:

If $f(x)$ is absolutely integrable on $\mathbb{R}$ and piecewise continuous on every finite interval, then the Fourier Transform $\hat{f}$ of f exists.

Linearity of Fourier Transform:

The Fourier Transform is a linear Operation: that is, for any functions f and g whose Fourier Transforms exist and constants a and b , the Fourier Transform of $af + bg$ exists and

$$\mathcal{F}(af + bg) = a\mathcal{F}(f) + b\mathcal{F}(g)$$

Proof:

$$\begin{aligned}\mathcal{F}(af + bg)(s) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (af(t) + bg(t)) e^{-ist} dt \\ &= a \times \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-ist} dt + b \times \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(t) e^{-ist} dt \\ &= a \mathcal{F}(f) + b \mathcal{F}(g)\end{aligned}$$

Ex: Let $f(x) = \begin{cases} k & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$

Compute the Fourier Transform.

$$\hat{f}(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ist} f(t) dt = \frac{1}{\sqrt{2\pi}} \int_{-1}^1 e^{-ist} \times k dt$$

$$= \frac{k}{\sqrt{2\pi}} \cdot \left. \frac{e^{-ist}}{(-is)} \right|_{-1}^1 = \frac{k}{\sqrt{2\pi}} \left[\frac{e^{is} - e^{-is}}{is} \right]$$

$$= \frac{k}{\sqrt{2\pi}} \frac{(\cos s + i \sin s) - (\cos s - i \sin s)}{is}$$

$$= \frac{k}{\sqrt{2\pi}} \frac{2i \sin s}{is} = \boxed{k \sqrt{\frac{2}{\pi}} \frac{\sin s}{s}}$$

$$\underline{\text{Ex:}} \quad \left. \begin{aligned} \text{Let } f(x) &= e^{kx} \text{ if } x < 0 \text{ where } k > 0 \\ &= 0 \text{ otherwise} \end{aligned} \right\}$$

Compute the Fourier Transform:

$$\begin{aligned} \hat{f}(s) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ist} f(t) dt = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^0 e^{-ist} e^{kt} dt \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^0 e^{(k-is)t} dt = \frac{1}{\sqrt{2\pi}} \left. \frac{e^{(k-is)t}}{(k-is)} \right|_{-\infty}^0 \\ &= \boxed{\frac{1}{\sqrt{2\pi} (k-is)}} \quad \left(\begin{aligned} &\left| e^{(k-is)t} \right| = \left| e^{-ist} e^{kt} \right| \\ &= e^{kt} \rightarrow 0 \text{ as } t \rightarrow -\infty \\ &\text{since } k > 0 \end{aligned} \right) \end{aligned}$$

Fourier Transform of the Derivative of $f(x)$:

Let f be continuous on $\mathbb{R}$ and $f(x) \rightarrow 0$ as $|x| \rightarrow \infty$. Furthermore let $f'(x)$ be absolutely integrable on $\mathbb{R}$.

$$\text{Then } \mathcal{F}(f')(s) = is \mathcal{F}(f)(s)$$

$$\begin{aligned} \underline{\text{Proof:}} \quad \mathcal{F}(f')(s) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f'(t) e^{-ist} dt \\ &= \frac{1}{\sqrt{2\pi}} \left[f(t) e^{-ist} \Big|_{-\infty}^{\infty} - (-is) \int_{-\infty}^{\infty} f(t) e^{-ist} dt \right] \end{aligned}$$

Since $f(t) \rightarrow 0$ as $|t| \rightarrow \infty$, we get

$$\boxed{\mathcal{F}(f')(s) = is \mathcal{F}(f)(s)}$$

• continuing in this way: $\mathcal{F}(f'')(s) = -s^2 \mathcal{F}(f)(s)$
 $\vdots$

$$\text{and } \boxed{\mathcal{F}(f^{(n)})(s) = (is)^n \mathcal{F}(f)(s)}$$

Ex: Find the Fourier Transform of $x e^{-x^2}$.

$$\mathcal{F}(x e^{-x^2})(s) = \mathcal{F}\left(-\frac{1}{2} (e^{-x^2})'\right)(s)$$

$$= -\frac{1}{2} \mathcal{F}\left((e^{-x^2})'\right)(s)$$

$$= -\frac{1}{2} is \mathcal{F}(e^{-x^2})(s)$$

$$= -\frac{is}{2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ist} e^{-t^2} dt$$

$$= \frac{-is}{2\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(t^2 + 2 \frac{is}{2} t + \left(\frac{is}{2}\right)^2\right)} e^{\left(\frac{is}{2}\right)^2} dt$$

$$= \frac{-is}{2\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(t + \frac{is}{2}\right)^2} e^{-\frac{s^2}{4}} dt$$

$$= e^{-s^2/4} \frac{(-is)}{2\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(t + \frac{is}{2}\right)^2} dt$$

$$= e^{-s^2/4} \frac{-is}{2\sqrt{2\pi}} \times \sqrt{\pi}$$

$$= \boxed{\frac{-is}{2\sqrt{2}} e^{-\frac{s^2}{4}}}$$

- In the above we have used the following result : $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$

Proof: Let $I = \int_{-\infty}^{\infty} e^{-x^2} dx$

$$\text{Then } I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right) \left(\int_{-\infty}^{\infty} e^{-y^2} dy \right)$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy$$

Let $x = r \cos \theta$, $y = r \sin \theta$; then $r^2 = x^2 + y^2$

and $dx dy = r dr d\theta$

$$\text{Then } I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta$$

$$= \int_0^{2\pi} \left[-\frac{e^{-r^2}}{2} \right]_0^\infty d\theta = \frac{1}{2} \times 2\pi = \pi$$

So, $\boxed{I = \sqrt{\pi}}$

Ex: Consider the Wave Equation

$$u_{tt} = c^2 u_{xx}$$

($u=u(x,t)$ is a function of x and t)

Taking the Fourier Transform of the equation in variable x

$$\frac{\partial^2 [\mathcal{F}(u)(s,t)]}{\partial t^2} = c^2 (-s^2) \mathcal{F}(u)(s,t)$$

$$\Rightarrow (\hat{u})_{tt} = -c^2 s^2 \hat{u} \quad \left(\begin{array}{l} \hat{u} \text{ is the Fourier} \\ \text{Transform in variable } x \end{array} \right)$$

Let $\hat{u} = y$

Then we have $y'' + c^2 s^2 y = 0$

$$\Rightarrow \hat{u}(s,t) = y = A(s) \cos(cst) + B(s) \sin(cst)$$

Taking the inverse Fourier Transform,

we will get

$$u(x, t) = \mathcal{F}^{-1}(\hat{u})(x, t).$$

Fourier Transform of the integral

$$\mathcal{F}\left\{\int_{-\infty}^t f(\tau) d\tau\right\} = \frac{\mathcal{F}(f)}{i\omega} + c\delta(\omega)$$

where c is a value such that

$$\int_{-\infty}^t (f(\tau) - c) d\tau = 0$$

It is normally referred to as the DC or average value.

Fourier Transform of the Convolution:

Convolution: The convolution $f * g$ of functions f and g is defined by

$$(f * g)(x) = \int_{-\infty}^{\infty} f(p) g(x-p) dp = \int_{-\infty}^{\infty} f(x-p) g(p) dp$$

Convolution Theorem:

Suppose that $f(x)$ and $g(x)$ are piecewise continuous, bounded and absolutely integrable on $\mathbb{R}$.

$$\text{Then } \mathcal{F}(f * g) = \sqrt{2\pi} \mathcal{F}(f) \mathcal{F}(g)$$

Proof:

$$\mathcal{F}(f * g)(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(p) g(t-p) dp \right) e^{-ist} dt$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} g(t-p) e^{-ist} dt \right) f(p) dp$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} g(q) e^{-is(p+q)} dq \right) f(p) dp$$

(let $t-p = q$)

$$= \left\{ \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-isp} f(p) dp \right\} \left\{ \int_{-\infty}^{\infty} g(q) e^{-isq} dq \right\}$$

$$= \sqrt{2\pi} \left\{ \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-isp} f(p) dp \right\} \left\{ \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-isq} g(q) dq \right\}$$

$$= \sqrt{2\pi} \mathcal{F}(f)(s) \mathcal{F}(g)(s)$$

$$= \sqrt{2\pi} (\mathcal{F}(f) \mathcal{F}(g))(s)$$

Hence $\boxed{\mathcal{F}(f * g) = \sqrt{2\pi} \mathcal{F}(f) \mathcal{F}(g)}$

Calculation of the convolution:

Since $\mathcal{F}(f * g)(s) = \sqrt{2\pi} \mathcal{F}(f)(s) \mathcal{F}(g)(s)$,
taking inverse Fourier Transform, we get

$$(f * g)(t) = \mathcal{F}^{-1}(\sqrt{2\pi} \hat{f}(s) \hat{g}(s))$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sqrt{2\pi} \hat{f}(s) \hat{g}(s) e^{ist} ds$$

So, $\boxed{(f * g)(t) = \int_{-\infty}^{\infty} \hat{f}(s) \hat{g}(s) e^{ist} ds}$ { This formula is useful in solving PDE